

Notes: Bikhchandani, Hirshleifer, and Welch (JPE, 1992)

A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades

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1 Big picture

- **Question.** Why do people often converge to the same behavior even when there are no obvious sanctions, payoff externalities, or direct taste for conformity? And why is that conformity often so fragile?
- **Setting.** Individuals act sequentially. Each person receives a private noisy signal about whether adopting some action is desirable, but observes only the *actions* of previous individuals rather than their private signals.
- **Core challenge.** Observing actions is much less informative than observing signals. Because actions are coarse, later agents may not be able to infer exactly what predecessors knew.
- **Main theoretical result.** In a fairly general sequential social-learning environment, an **informational cascade** almost surely starts: at some point, an individual finds it optimal to ignore his own private information and simply follow the pattern created by predecessors.
- **Mistake result.** Cascades can be **wrong**. Society may converge to a bad action because only a little early information gets aggregated before later decisions become uninformative.
- **Fragility result.** Cascades are brittle. Once a cascade starts, later followers do not add information, so even a small public information release or even the mere possibility of a change in fundamentals can overturn long-lived conformity.
- **Fads result.** Because cascades rest on thin informational foundations, convention changes can happen more often than underlying fundamentals actually change. This gives a theory of fads, fashions, booms, and crashes.
- **Why the paper matters.** This is a foundational theory of herding and social learning. Its key contribution is not just to explain conformity, but to explain *error-prone* conformity and abrupt reversals.

2 Data and measurement (key objects)

This is a **theory paper**; there is no empirical dataset. The relevant objects are private signals, observed actions, posterior beliefs, and the timing of decisions.

2.1 Agents, actions, and sequential timing

- There is a sequence of individuals $i = 1, 2, \dots$
- Each individual chooses one of two actions: **adopt** or **reject**.
- Individual i observes the actions of all predecessors, but not their private signals.
- The ordering of moves is exogenous and common knowledge.

2.2 Benchmark binary objects

In the benchmark example:

- the gain from adoption is a common value

$$V \in \{0, 1\}, \quad \Pr(V = 1) = \Pr(V = 0) = \frac{1}{2};$$

- the cost of adoption is common and initially fixed at

$$C = \frac{1}{2};$$

- each individual observes a private signal

$$X_i \in \{H, L\};$$

- the signal is informative:

$$\Pr(X_i = H \mid V = 1) = p_i > \frac{1}{2}, \quad \Pr(X_i = H \mid V = 0) = 1 - p_i.$$

When all individuals have the same precision, the paper writes $p_i = p$ for all i .

2.3 General model objects

The general model allows:

- a finite set of adoption values

$$V \in \{v_1, v_2, \dots, v_S\}, \quad v_1 < \dots < v_S;$$

- a finite signal space

$$X_i \in \{x_1, x_2, \dots, x_R\}, \quad x_1 < \dots < x_R;$$

- conditional signal probabilities

$$P_{q\ell} = \Pr(X_i = x_q \mid V = v_\ell);$$

- an action history

$$A_i = (a_1, a_2, \dots, a_i).$$

2.4 Information sets and decision-relevant objects

- For a given history and action, $J_i(A_{i-1}, a_i)$ denotes the set of signal realizations that would lead individual i to choose action a_i .
- The crucial object is the posterior expected value of adoption conditional on one's own signal and on the observed history of past actions.
- A cascade occurs when this posterior makes the individual's action independent of his private signal.

2.5 Unobservables and timing extensions

- The basic model keeps the underlying value fixed.
- Later sections allow for **public information releases** and for **changes in the underlying value**.
- In the fads example, the underlying value can shift at a particular date, creating the possibility that social behavior moves even more frequently than the fundamentals do.

2.6 Conceptual applications

The paper applies the cascade logic to:

- politics and election momentum,
- zoology and territorial clustering or mate copying,
- medical practice and scientific theory adoption,
- finance, debt renegotiation, IPO subscription, and bank runs,
- peer influence and stigma.

3 Models and empirical specifications (all equations + interpretation)

3.1 Specific binary model

The benchmark model is designed to isolate the mechanism as clearly as possible. The environment is:

$$\begin{aligned} V &\in \{0, 1\}, & \Pr(V = 1) &= \Pr(V = 0) = \frac{1}{2}, \\ C &= \frac{1}{2}, \\ X_i &\in \{H, L\}, \end{aligned}$$

with signal structure

$$\Pr(X_i = H \mid V = 1) = p_i > \frac{1}{2}, \quad \Pr(X_i = H \mid V = 0) = 1 - p_i. \quad (1)$$

- **Interpretation.** Adoption is worthwhile if the posterior probability that $V = 1$ is high enough.
- **Social-learning friction.** Individuals do not observe others' signals; they see only whether predecessors adopted or rejected.

3.2 Decision rule in the symmetric binary case

Suppose $p_i = p$ for all i .

- Individual 1 adopts if he observes H and rejects if he observes L .

- Individual 2 can infer individual 1's signal from the observed action. If the first individual adopted, then the second adopts for sure after seeing H , and is indifferent after seeing L .
- Symmetrically, if the first individual rejected, the second rejects for sure after seeing L , and is indifferent after seeing H .
- By the third individual, there are three cases: two adoptions, two rejections, or one of each. Two identical actions start a cascade; mixed actions reset the problem.
- **Interpretation.** The key feature is that the first two actions can compress all the relevant social information into a very small amount of evidence.
- **Why this matters.** Once the third individual ignores his own signal, all later individuals do so as well.

3.3 Cascade probabilities in the benchmark model

In the symmetric binary case, after two individuals the unconditional probabilities of an up cascade, no cascade, and a down cascade are

$$\Pr(\text{UP after 2}) = \Pr(\text{DOWN after 2}) = \frac{1 - p(1 - p)}{2} = \frac{1 - p + p^2}{2}, \quad (2)$$

$$\Pr(\text{no cascade after 2}) = p(1 - p). \quad (3)$$

After an even number of individuals n , the probability that no cascade has yet begun is

$$\Pr(\text{no cascade after } n) = [p(1 - p)]^{n/2}, \quad (4)$$

so the probability of being in either an up or down cascade is

$$\Pr(\text{UP after } n) = \Pr(\text{DOWN after } n) = \frac{1 - [p(1 - p)]^{n/2}}{2}. \quad (5)$$

- **Interpretation.** The probability of not yet being in a cascade falls geometrically, so convergence to conformity is very fast.
- **Economic point.** More precise private information (higher p) makes a cascade begin sooner.

3.4 Correct and incorrect cascades

Conditioning on the true state $V = 1$, after two individuals the probabilities of a correct up cascade, no cascade, and a wrong down cascade are

$$\Pr(\text{UP after 2} \mid V = 1) = \frac{p(p + 1)}{2}, \quad (6)$$

$$\Pr(\text{no cascade after 2} \mid V = 1) = p(1 - p), \quad (7)$$

$$\Pr(\text{DOWN after 2} \mid V = 1) = \frac{(2 - p)(1 - p)}{2}. \quad (8)$$

For even n , these become

$$\Pr(\text{UP after } n \mid V = 1) = \frac{p(p+1)}{2(1-p+p^2)} \left[1 - (p(1-p))^{n/2} \right], \quad (9)$$

$$\Pr(\text{DOWN after } n \mid V = 1) = \frac{(2-p)(1-p)}{2(1-p+p^2)} \left[1 - (p(1-p))^{n/2} \right]. \quad (10)$$

- **Interpretation.** Even when signals are informative, the wrong cascade can still occur with nontrivial probability.
- **Core mechanism.** Once the system falls into a cascade, later signals are no longer aggregated. So society can end up locking onto the wrong action based on little early evidence.

3.5 General model and definition of an informational cascade

In the general setup, individual $n+1$ observes his own signal realization x_q and the action history $A_n = (a_1, \dots, a_n)$. Let $J_i(A_{i-1}, a_i)$ be the set of signal realizations consistent with individual i having taken action a_i after history A_{i-1} . Then the posterior expected value of adoption is

$$V_{n+1}(x_q; A_n) = E \left[V \mid X_{n+1} = x_q, X_i \in J_i(A_{i-1}, a_i) \text{ for all } i \leq n \right]. \quad (11)$$

Individual $n+1$ adopts if

$$V_{n+1}(x_q; A_n) \geq C. \quad (12)$$

The paper's formal definition is:

Definition. An *informational cascade* occurs if an individual's action does not depend on his private information signal.

- **Interpretation.** The action rule becomes flat in the private signal.
- **Immediate implication.** If individual i is in a cascade, then his action conveys no new information. Since later individuals observe only actions, not signals, everyone after i is also in a cascade.

3.6 Assumptions behind the general result

The paper imposes two key regularity conditions.

Assumption 1: Monotone likelihood ratio ordering. For all signal values $q < R$ and states $\ell < S$,

$$\frac{P_{q\ell}}{P_{q,\ell+1}} \geq \frac{P_{q+1,\ell}}{P_{q+1,\ell+1}}, \quad (13)$$

with strict inequality for at least one q .

Assumption 2: No long-run ties.

$$v_\ell \neq C \quad \text{for all } \ell. \quad (14)$$

- **Interpretation of Assumption 1.** Higher signal realizations make higher values of adoption more likely.
- **Interpretation of Assumption 2.** When enough information accumulates, agents are not asymptotically indifferent between adopting and rejecting.

3.7 Main positive result: a cascade almost surely starts

Under the two assumptions above, the paper proves:

Proposition 1. If assumptions 1 and 2 hold, then as the number of individuals increases, the probability that a cascade eventually starts approaches one.

- **Interpretation.** If a long sequence passes without a cascade, then many predecessor actions must still be informative. But then the law of large numbers implies that history alone nearly reveals the true value, making the next individual's own signal negligible. That individual therefore ignores his signal and starts a cascade.
- **Economic meaning.** Conformity is not a knife-edge event. In the model it is a generic long-run outcome.

3.8 Mistaken cascades and the loss of information aggregation

The fact that cascades start does *not* mean that social learning works well.

- Once a cascade begins, later actions stop transmitting private information.
- Hence society may end up relying on the information of only a few early movers.
- Figure 1 in the paper makes this vivid: even when the signal is fairly accurate, the probability of an incorrect cascade remains economically important.
- **Interpretation.** The problem is not that people behave irrationally. The problem is that rational inference from others' *actions* can shut down later information aggregation.

3.9 Fashion leaders and heterogeneous precision

The paper next studies the case in which signal precisions differ across individuals.

Result 1. In the binary signal case, if the highest-precision individual decides first, then that individual's decision is followed by all later individuals. Moreover, when all later individuals have identical precision, they are better off if the first individual's precision is slightly *lower* rather than slightly *higher* than theirs.

- **Interpretation.** A very well-informed first mover can become a **fashion leader**. His action is so informative that everyone else rationally defers immediately.
- **Second part of the result.** This immediate deference is socially costly because it suppresses later information aggregation. If the first mover is just a bit less dominant, later decisions incorporate more independent information.

3.10 Public information and fragility

The paper studies how public information interacts with cascades.

Result 2. The release of public information before the first individual’s decision can make some later individuals worse off in an ex ante sense.

- **Interpretation.** Public information has a direct positive effect, but it can also distort the informational content of early actions. Very noisy public information can therefore reduce later welfare.

After a cascade has started, however, the conclusion changes.

Proposition 2. If all individuals’ signals are drawn from the same distribution, then after a cascade has begun, all individuals welcome public information.

- **Interpretation.** Once a cascade is under way, subsequent private actions convey no further information anyway. Public information then helps directly without crowding out informative actions.

The fragility result is even stronger:

Result 3. The release of a small amount of public information can shatter a long-lasting cascade, where “small” means less informative than the private signal of a single individual.

- **Interpretation.** The public signal does not need to outweigh the information of all later imitators. It only needs to offset the thin informational margin on which the cascade originally started.

Finally, repeated public information eventually restores correct social learning.

Proposition 3. If, before each individual chooses, there is a probability bounded away from zero of a public information release, and these releases are conditionally i.i.d., then individuals eventually settle into the correct cascade.

- **Interpretation.** Repeated public information keeps adding genuine evidence, so eventually the posterior concentrates on the true value.
- **Key contrast.** Without new information, wrong cascades can persist forever; with occasional public information, they are ultimately corrected.

3.11 Fads: behavior can change more often than fundamentals

The paper’s final formal extension allows the underlying value to change at a later date. In the numerical example:

- up to individual 100, the value is the original V ;
- at individual 101, with probability 0.1, the true value is redrawn;

- therefore the post-change value W satisfies

$$\Pr(W = V) = 0.95, \quad \Pr(W = 1 - V) = 0.05. \quad (15)$$

Under full information, or when previous signals are observable, behavior changes with probability approximately 0.05, matching the true probability of a value change.

But when only previous actions are observable, the paper proves:

Result 4. In the numerical fads example, the probability of a cascade reversal exceeds 0.0935, even though the true probability that the correct action changes is only 0.05.

- **Interpretation.** Social conventions can swing almost twice as often as fundamentals do.
- **Big idea.** This is a theory of fads: abrupt social reversals need not reflect large new shocks. They can arise because the old convention rested on weak information.

4 Identification and instruments (what makes the estimates credible)

This is a **theory** paper, so there are no econometric instruments. The relevant question is instead: *what assumptions make the cascade mechanism work, and why is the mechanism distinct from other conformity stories?*

4.1 What fails in standard explanations of conformity

The paper starts by contrasting informational cascades with four common explanations of social uniformity:

- sanctions on deviants,
- payoff externalities,
- direct preference for conformity,
- communication.
- **Why these are not enough.** Sanctions, payoff externalities, and conformity preference can explain uniform behavior, but they usually imply that conformity becomes *more robust* as more people join. Costless and credible communication, in contrast, should lead society toward the correct outcome rather than persistent mistakes.
- **The gap.** These theories do not naturally explain *fragile*, error-prone mass behavior.

4.2 What generates cascades in the new approach

The mechanism requires a particular informational structure:

- decisions are sequential,
- each individual has private information,

- later individuals observe only earlier *actions*, not signals,
- actions are coarse enough that they do not fully reveal the signals that generated them.
- **Interpretation.** This is the core identification logic of the model. If later agents saw predecessors' signals instead of their actions, information would continue to accumulate correctly and mistaken cascades would be much less likely.

4.3 Why the monotone likelihood ratio property matters

Assumption 1 guarantees that higher signals systematically tilt beliefs toward higher adoption values.

- **Role in the proof.** It makes posterior beliefs monotone in the signal realization, so the action rules have the threshold structure needed for the cascade argument.
- **Economic meaning.** Signals can be ranked from “bad” to “good” in a consistent way.

4.4 Why the no-long-run-ties assumption matters

Assumption 2 rules out the possibility that even with a lot of information agents remain exactly indifferent.

- **Role in the proof.** Once the history becomes sufficiently informative, the agent strictly prefers one action, which lets a cascade start.
- **Without it.** The history could converge to a knife-edge case where the next agent's own signal still matters forever.

4.5 Why fragility is intrinsic rather than accidental

The fragility result is not driven by society sitting accidentally near a boundary.

- In sanction or payoff-externality models, large social reversals occur only if the system happens to be near the tipping point.
- In this paper, society *systematically* ends up near the boundary because the cascade stops information aggregation early.
- Hence small shocks, or even the possibility of value change, can overturn mass behavior.

4.6 Relation to the literature

- Relative to earlier conformity theories, the paper's distinctive contribution is the explanation of **fragile** conformity.
- Relative to Banerjee and Welch, the paper emphasizes how cascades can explain not only herding, but also **rapid social reversals** such as fads.
- Relative to models with communication, the paper stresses that action-based observation is informationally coarse and therefore can produce systematic errors.

TABLE 1
SIGNAL PROBABILITIES

	$\Pr(X_i = H V)$	$\Pr(X_i = L V)$
$V = 1$	p_i	$1 - p_i$
$V = 0$	$1 - p_i$	p_i

5 Tables and figures

5.1 Table 1: Signal probabilities

Read:

- Table 1 defines the benchmark binary signal structure: a high signal is more likely when adoption is truly valuable, and a low signal is more likely otherwise.
- The whole benchmark example turns on the precision parameter p .
- When p is close to $1/2$, private signals are noisy, and wrong cascades become more likely.

5.2 Equations (2)–(10) and Figure 1: Cascades start quickly, but can be wrong

Read:

- Equation (4) shows that the probability of not yet being in a cascade falls geometrically in the number of decision makers.
- Figure 1 shows that increasing signal precision raises the probability of a correct cascade, but the probability of an incorrect cascade remains sizable even for informative signals.
- The visual message is that social convergence is fast, but not necessarily accurate.

5.3 Result 1: Fashion leaders

Read:

- Small differences in first-mover precision can dramatically change the path of social learning.
- If the best-informed person moves first, later individuals rationally imitate immediately.
- This helps explain why a few early opinion leaders can have disproportionate influence.

5.4 Result 2, Proposition 2, and Result 3: Public information

Read:

- Noisy public information introduced *before* imitation starts can reduce welfare by distorting how later observers interpret early actions.

TABLE 2
EFFECT OF MULTIPLE INFORMATION RELEASES

p	NO PUBLIC INFORMATION			PUBLIC INFORMATION (1)			PUBLIC INFORMATION (10)		
	Cascade		Expectation of Difference (3)	Cascade		Expectation of Difference (6)	Cascade		Expectation of Difference (9)
	Up (1)	Down (2)		Up (4)	Down (5)		Up (7)	Down (8)	
.55	.5645	.4355	.262	.5857	.4132	.451	.6767	.3200	1.691
.65	.6950	.3050	.783	.7396	.2601	1.236	.8986	.1005	4.577
.75	.8100	.1900	1.242	.8633	.1366	1.950	.9826	.0172	6.874
.85	.9065	.0935	1.623	.9480	.0516	2.494	.9989	.0011	8.940
.95	.9724	.0276	1.890	.9884	.0116	2.858	1.0000	.0000	10.982

- Once a cascade exists, public information is always good because it is the only new information entering the system.
- Remarkably, a very small public signal can overturn a very long-lived cascade.

5.5 Table 2: Repeated public releases make the correct cascade much more likely

Read:

- Table 2 reports a numerical illustration for the binary example with occasional public information.
- Even rare public releases sharply raise the probability that society is in the correct cascade by the time the 1,000th individual acts.
- For example, when $p = .75$, the probability of ending in the correct cascade rises from about 0.81 with no public releases to about 0.86 with one release per 1,000 individuals and about 0.98 with ten releases per 1,000 individuals.

5.6 Section IV examples: where the mechanism might matter

Read:

- The paper deliberately applies the same logic across politics, animal behavior, medical practice, finance, peer effects, and stigma.
- The common pattern is sequential decision-making under uncertainty where people infer information from *what others do*.
- These examples are not formal tests; they are meant to show the broad scope of the mechanism.

5.7 Result 4 and Appendix tables A1–A5: Fads can exceed fundamental change

Read:

- The appendix constructs numerical examples showing both that noisy public information can hurt initially and that conventions can reverse more often than true values change.
- In the fads example, the probability of convention reversal is greater than 0.0935, whereas the probability that the true underlying value changes is only 0.05.
- The paper’s key message is therefore stronger than “society can make mistakes.” It is that social behavior can move too much relative to fundamentals.

TABLE A1
CONDITIONAL SIGNAL PROBABILITY

	x_1	x_2	x_3
$\Pr(X_i V = 0)$	.4	.55	.05
$\Pr(X_i V = 1)$	.2	.7	.1

TABLE A2
THE DECISION OF INDIVIDUAL 1 WITHOUT PUBLIC INFORMATION RELEASE

	Unconditional Probability (1)	Expected Value (2)	Action (3)
X_1			
x_1	.3000	.3333	reject
x_2	.6250	.5600	adopt
x_3	.0750	.6667	adopt

TABLE A3
THE DECISION OF INDIVIDUAL 2 WITHOUT PUBLIC INFORMATION RELEASE

D_1, X_2	Unconditional Probability (1)	Expected Value (2)	Action (3)
R, x_1	.1000	.2000	reject
R, x_2	.1800	.3839	reject
R, x_3	.0200	.5000	reject
A, x_1	.2000	.4000	reject
A, x_2	.4450	.6292	adopt
A, x_3	.0550	.7273	adopt

TABLE A4
THE DECISION OF INDIVIDUAL 1 WITH PUBLIC INFORMATION RELEASE

S, X_1	Unconditional Probability (1)	Expected Value (2)	Action (3)
L, x_1	.151	.3245	reject
L, x_2	.31175	.55012	reject
L, x_3	.03725	.657718	adopt
H, x_1	.149	.34228	reject
H, x_2	.31325	.5698	adopt
H, x_3	.03775	.6755	adopt

TABLE A5
THE DECISION OF INDIVIDUAL 2 WITH PUBLIC INFORMATION RELEASE

S, D_1, X_2	Unconditional Probability (1)	Expected Value (2)	Action (3)
L, A, x_1	.01	.49	reject
L, A, x_2	.0242	.71	adopt
L, A, x_3	.0031	.794	adopt
L, R, x_1	.14176	.313	reject
L, R, x_2	.2847	.537	reject
L, R, x_3	.0339	.6454	adopt
H, A, x_1	.0996	.4096	reject
H, A, x_2	.22365	.638	adopt
H, A, x_3	.02775	.735	adopt
H, R, x_1	.0494	.2065	reject
H, R, x_2	.0896	.3984	reject
H, R, x_3	.01	.5100	reject

6 Short summary

- **Conclusion.** Sequential social learning based on observed actions can generate conformity, but that conformity may be based on very little information.
- **Main theoretical insight.** Because actions are a coarse signal of predecessors' private information, later individuals can rationally ignore their own signals and herd.
- **Main implication.** Cascades explain not just localized conformity but also fragility, reversals, and fads.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that rational individuals can end up in mass conformity without any direct taste for conformity and without strong payoff externalities.
- It shows that such conformity can begin quickly and can be wrong.
- It also shows that conformity generated in this way is fundamentally fragile because later imitators do not deepen the information base.

7.2 Economic intuition

The core intuition is simple: if people observe only the decisions of others, then an action compresses a lot of information into a coarse binary message. Early on, this can be useful because later individuals

learn something from predecessors. But once enough people have taken the same action, later agents may rationally infer that the social evidence already outweighs their own signal. At that point, they stop using their own information. The learning process then shuts down. Society converges, but convergence is based on a very thin informational foundation.

That thin foundation is why cascades are brittle. One small piece of new public evidence, or even some suspicion that the environment has changed, can be enough to make people start listening to their private information again. The same social system that converged rapidly can therefore reverse rapidly.

7.3 Takeaway

This paper explains why societies can look simultaneously *conformist* and *unstable*. Informational cascades make it rational to follow the crowd, but because the crowd's behavior may embody only a little real information, the resulting social order is precarious. That is why the same mechanism can generate localized norms, mistaken conventions, sudden fashion shifts, booms, crashes, and fads.